

# **Non-local quantum evolution of entangled ensemble states in neural nets and its significance for brain function and a theory of consciousness**

Erhard Bieberich

Medical College of Virginia Campus of Virginia Commonwealth University

Department of Biochemistry and Molecular Biophysics

1101 E. Marshall St. P.O.Box 614, Richmond, VA 23298-614

First version June 2<sup>nd</sup>, 1999, revised July 26<sup>th</sup>, 1999

## **Abstract**

Current quantum theories of consciousness suggest a configuration space of an entangled ensemble state as global work space for conscious experience. This study will describe a procedure for adjustment of the singlet evolution of a quantum computation to a classical signal input by action potentials. The computational output of an entangled state in a single neuron will be selected in a network environment by "survival of the fittest" coupling with other neurons. Darwinian evolution of this coupling will result in a binding of action potentials to a convoluted orbit of phase-locked oscillations with harmonic, m-adic, or fractal periodicity. Progressive integration of signal inputs will evolve a present memory space independent from the history of construction. Implications for mental processes, e.g., associative memory, creativity, and consciousness will be discussed. A model for generation of quantum coherence in a single neuron will be suggested.

Introduction

Several features of human consciousness persistently elude scientific explanation. The emergence of a consciously observed world from a neuronal substrate entails a serious topological paradox: neuronal elements can be spatially separated whereas the space in which conscious perception takes place is experienced as an inseparable unity [1]. Any operation mapping the world experienced in consciousness onto a neuronal substrate will inevitably destroy the integrity of the self as inner observer by spatial distribution resulting in separated and reduced sub-selves. More intuitively than rigorously explainable by scientific reasoning, a solution of this paradox has been approached by consciousness models based on a configuration space arising from non-local quantum effects in the brain [2,3]. In particular, effects based on quantum coherence or non-locality, e.g., EPR (Einstein-Podolsky-Rosen) phenomena or entanglement have been suggested to provide mechanisms by which the self stays holistic and irreducible despite of being spatially distributed.

According to the most advanced model it is assumed that elements of the cytoskeleton, so called microtubules, form a neuronal substrate for quantum computation by entangled electron-singlet states which are caged in the tubular lumen shielded from environmental heat pollution [4,5]. Most recently, this and other models of consciousness have been criticized due to extremely short decoherence times presumably rendering quantum computation impossible in thermally equilibrated brain tissue [6]. On the other hand, it should not be neglected that any classical model developed so far does only account for localized neuronal computation, but provides no concept for non-local inner observation as inevitable for consciousness. The preference of quantum computation for a description of the actual event of conscious experience over classical neuronal computation emerges primarily from the concept of non-locality inherent to quantum coherence and the self as inner observer. However, even considered quantum coherence in the brain to be technically difficult but realizable, it remains enigmatic how: i), any quantum-computational model for brain function and consciousness will integrate in- and output by classical signal transduction in form of synaptic action potentials, and ii), a space unity for conscious experience will be formed by participation of entangled quantum states on separated neurons. It is hard to imagine how classical transduction channels as in i) can simultaneously feed the variety of visually, acoustically, emotionally etc. experienced information into a single neuron. Instead, it is suggested that a set of separated quantum coherent states in clustered neurons may form a common configuration space by sharing entanglement or EPR. It will be attempted to bridge an explanatory gap between orthodox neuronal and quantum computation by the use of basic concepts from both disciplines.

In the first part of this study it will be shown that two coupled neurons can form a cNOT-gate by communicating one qubit by two classical channels (axon/dendrite connections). The respective algorithms and operations will form the basis for the concept of neural quantum computation and classical communication (NQCC). We will then discuss how a complete register of qubits can be decompressed by multiple classical channels with the potential of a probabilistic quantum computation and superselection by Darwinian evolution. In the third section of the first part we will introduce the concept of quantum oscillation

in a single neuron as intrinsic dynamics of the unitary evolution and the measurement process. This concept will then be correlated with the multiple channel model for superselection of a particular quantum oscillation. The second part of the study is dedicated to the significance of NQCC to higher mental function. It will be shown that Darwinian evolution of quantum oscillation can stabilize orbits for a memory space consistent with current models developed from orthodox neuronal computation. In the third part we will derive a model for the generation of consciousness in a potential biological substrate.

## Part 1: Decompression and Darwinian evolution of quantum information in neural networks

### 1.1. Basic rules for neural quantum computation and classical communication (NQCC)

The following model of neural quantum operation and classical communication (NQCC) utilizes concepts from genetic evolution theory and quantum computation, and is a context-dependent application of local quantum operations and classical communication (LQCC) on neural networks. Due to the interdisciplinary character of this model the terminology will be "bilingually" with reference to both disciplines. Suppose, there is a system of qubits entangled in a pure state on a single neuron and evolving by unitary operation. By unknown intrinsic dynamics and interaction with its neuronal environment this system decoheres and communicates its computational result to classical channels. The unitary operation on the ensemble of qubits will be termed "mutation" if a coherent quantum state evolves from  $[A]$  to  $[A']$  on neuron 1 or  $[B]$  to  $[B']$  on neuron 2. In the simplest case  $[A]$ ,  $[A']$ ,  $[B]$ , and  $[B']$  can be 0 or 1 if only one qubit is measured. The mutation denotes a NOT-operation and the neuronal quantum state the genotype which is pure if not measured. The phenotypic manifestation of this genotype proceeds by measurement of the system of entangled qubits due to interaction with its neuronal environment. This measurement is achieved by "translation" of  $[A \leftrightarrow a]$ ,  $[A' \leftrightarrow a']$ ,  $[B \leftrightarrow b]$ , and  $[B' \leftrightarrow b']$ . Note, that the "translator" is a set of operators which is composed of "readout" by measurement and arbitrary "reset" of entanglement. Accordingly, the quantum states  $[A]$ ,  $[A']$ ,  $[B]$ , and  $[B']$  are eigenstates of the system whereas  $[a]$ ,  $[a']$ ,  $[b]$ , and  $[b']$  are pointer states of the neuronal environment. The correlation of the two ensembles of states is reversible and deterministic. The translator resets entanglement by Hadamard operation or qubit-rotation. It does not violate the "no cloning" theorem: the unitary operation of the system evolves by intrinsic dynamics, readout and reset are only for orthogonal states. The pointer states  $[a]$ ,  $[a']$ ,  $[b]$ , and  $[b']$  will then be converted into classical communication channels. This "conversion" is operationally complete:  $[a \leftrightarrow b]$ ,  $[a' \leftrightarrow b']$ ,  $[a \leftrightarrow b']$ , and  $[a' \leftrightarrow b]$ . It can be used for input programming and output reading of the qubit register on neuron 1 or 2 by classical neuronal computation. This implicates that the conversion is another operation composed of two parts: The communication of one qubit by two classical channels and the exchange of the results of qubit-registers (swapping) for the pointer-states.

The operator pairs "mutator" and "translator" can be localized on a single or on two spatially separated neurons. No matter whether the communication between the quantum states is managed by a quantum or a classical channel the quantum computational result will be identical if the following constellation holds: Suppose that neuron 1 does not mutate and stays in state  $[A] = 1$ . It will communicate  $[a \leftrightarrow b]$  to neuron 2 and its quantum register will be reset to  $[B] = 1$ . If neuron 2 mutates to  $[B'] = 0$  by a NOT-operation it will communicate  $[b' \leftrightarrow a]$  back to neuron 1, thus resulting in reset of  $[A]$  to 1. This sequence is reversible and can be run in both directions. It is identical to a controlled Not-gate (cNOT) with control neuron 1 and target neuron 2 [7,8]. The computational result will be indistinguishable from the communication between the system states A and B by quantum channels on a single neuron. However, it is obvious that the constellation of communication channels was arbitrarily chosen in order to make the model fit to the entanglement case. It will now be evaluated, how the system can intrinsically choose communication channels for purification of entanglement.

## 1.2. Purification of entanglement by "choice of channel"

A neuronal network can filter and select the coupling of quantum states at the moment of translation by a "choice of channel" procedure as follows: Suppose that there is no mutation on control neuron 1 and  $[A \leftrightarrow A] = 1$  whereas that on target neuron 2  $[B] = 0$  or 1 may take place  $[B \leftrightarrow B']$  or not  $[B \leftrightarrow B]$ . Note, that it is not assumed that  $[B]$  necessarily mutates to  $[B']$  as in case of a cNOT gate. The two neurons communicate by four classical channels performing the conversions  $[a \leftrightarrow b]$ ,  $[a' \leftrightarrow b']$ ,  $[a \leftrightarrow b']$ , and  $[a' \leftrightarrow b]$  at the same time. The signal from neuron 2 back to 1 may then take two of the four channels depending on the output of 2 (e.g., if there is mutation  $[B \leftrightarrow B']$  then the channels  $[a \leftrightarrow b']$  or  $[a' \leftrightarrow b]$  can be taken). The simultaneous communication on two different cbits-channels, however, will violate the deterministic correlation of the actual quantum state on neuron 1 with a particular classical channel (that is to say: there is either translation of  $[a]$  to  $[A]$  or  $[a']$  to  $[A']$ ). In order to allow a "choice of channel" by neuron 1, we introduce a feedback-loop operation by first measuring the system state on 1, e.g.  $[a]$  by translation of  $[A]$ . Then we allow a reset of  $[A]$  using  $[a]$  as input. This input, however, can also be obtained from neuron 2 by a classical AND-gate comparing output  $[a]$  on neuron 1 with input from the converter channels  $[a]$  or  $[a']$ . In case that  $[A] = 1$  and channel  $[a \leftrightarrow b']$  is chosen, the computational result is equivalent to that of a cNOT-quantum gate as described before. Note, that coupling is achieved by a NOT-operation of a quantum register on neuron 2 and a classical AND-operation depending on the result of the quantum register of neuron 1. The critical operation is given by the translation  $[A \leftrightarrow a]$  which includes a measurement of orthogonal system states that must be fully reversible by the reset operation. Recent models assume a realistic programming of entangled states by making particular spins addressable due to deterministic coupling to classical states, e.g. by cellular automaton architecture in a quantum dot lattice modulated by classical information channels [9-16]. The shortcoming of the model in its preliminary form is that the neurons are not coupled by a cNOT-operation if  $[B]$  does not mutate and channel  $[a \leftrightarrow b]$  is chosen.

In order to establish a choice of channel for a cNOT-operation at each point of time, the model is refined to a "choice of channel is choice of neuron" model. Again, we start with a condition  $[A \leftrightarrow A] = 1$  on neuron 1. The signal conversion  $[a \leftrightarrow b]$  is now communicated to two target neurons 2 and 2'. These target neurons communicate back to 1 on two of four channels as described before. In this case, however, there is an increased chance that there is a mutation of  $[B \leftrightarrow B']$ ,  $p = 0.75$  instead of  $p = 0.5$ . Note, the more neurons participate the higher is the chance to find at least one mutation ( $p = 1 - 0.5^n$  with  $n$  = number of target neurons). Decompression of the system state  $[A]$  will generate a population of ancilla for the pointer-states on neuron 1 by communication to a large number of coupled neurons. However, this redundancy, as beneficial it may be for decompression of quantum information, is unsatisfactory if a local measurement on a single neuron cannot decide which neuron is chosen for tracing out useful ancilla. In order to find a deterministic solution for a signal path establishing a deterministic instead of a probabilistic coupling we consider control and target neurons equivalent, that is to say: both neurons can mutate,  $[A \leftrightarrow A']$  and  $[B \leftrightarrow B']$ . One of two chosen channels  $[a' \leftrightarrow b]$  or  $[a' \leftrightarrow b']$  will only be taken when corresponding to an actual mutation  $[B \leftrightarrow B']$  on the neuron 2. At first sight, this is prohibited since the arbitrary determination of this condition undoes the freedom of neuron 2 for a realistic quantum decision. However, the condition "mutation is mandatory for coupling" will make sense if we increase the number of orthogonal system states on each neuron as potential result of a mutation. Reversibility of the computation is then only possible if a particular mutation on neuron 2 fits to that of neuron 1 by choice of the correct signal path. This "superselection" rule can trace out stable from probabilistic couplings of neurons by self-organization following Darwinian type evolution (see section below).

### 1.3. Induced fit and phase locking of coupled quantum oscillators

Suppose, that neuron 1 is coupled to as many neurons  $n$  as there are orthogonal output states of a quantum operation ( $n = 2^q$  with  $q$  = number of qubits). For simplicity of the model, it is assumed that i), there are four quantum states:  $A, A', A'', A'''$  on neuron 1, and  $B, B', B'', B'''$  on neuron  $N$  out of  $n$  possible, and ii), that each out of four neurons  $N = 2, 2', 2''$  and  $2'''$  oscillates between the ground state  $[B]$  and only one mutator state of the system, e.g.  $2 [B \leftrightarrow B], 2' [B \leftrightarrow B'], 2'' [B \leftrightarrow B''],$  and  $2''' [B \leftrightarrow B''']$ . The term quantum oscillation will be used in its broadest sense, as periodic fluctuation of the unitary operator between two pure states of entangled qubits which follows intrinsic (unknown) dynamics, however, with orthogonal eigenstates corresponding to those of the pointer states in the neuronal environment. In its narrower sense it may also be used as harmonic motion of quantum coherent particles [17]. These premises appear to be arbitrary, but that does not affect the functional validity of the model. Quantum state  $[A]$  will now be prepared in neuron 1 and will oscillate between the quantum states  $[A'], [A''],$  or  $[A''']$  with different phases. Provided that the phase of a mutator oscillation in neuron 1 must fit to that of neuron 2, 2', 2'', or 2''' in order not to be eliminated, there will be a specific mutation in neuron 1 selected by "induced fit" and a certain target neuron coupled by phase-locking according to the "choice of channel is choice of neuron" model. Note, that only the first choice of mutation in neuron 1 is free by unitary operation, and the phase is then interlocked by a feedback-loop

operation with a particular channel from neuron N as described in Part 1.2. This model can easily be generalized to a larger number of interlocked neurons purifying a certain phase of network oscillation. Moreover, parts of the network will form resonance states with its environment thereby providing a rich source of associative memory. The significance of superselected quantum oscillations for higher mental function will be discussed in the following part of the study.

## Part 2: The significance of neural quantum computation for function and biology of the brain

### 2.1. Decompression of qubits by tuning of cbits

The input of classical information channels is able to select or phase-lock a particular oscillation by iterative input/output feedback-loops on a quantum operation as described previously. A cNOT-gate may thus drive all of the allowed transitions, but is only resonant with one of them. Suppose now, that i), a coherent neuronal substrate oscillates between  $f=2^q$  ( $f$ =number of input/output channels,  $q$ =number of qubits) orthogonal sub-system (eigen)states, and ii), the quantum oscillation is connected to the classical information channels of  $n$  control or feeder neurons. Even if  $f$  is as small as  $1/\text{neuron}$  ( $f=n$ ), the time difference between input and output-channel transmission will be able to phase-lock an immense rich scale of oscillations or memory states in the target neuron. At this point, it should be noted that networking of neurons in the human brain efficiently provides the architecture required for coupling as described in this Letter. Each neuron in the cortical layer comprises convergent dendritic input and divergent axonic output channels which establish a connection with an average number of more than  $10^4$  different neurons in a network with tremendous feedback potential [18-21]. In a simplified scenario, the time delay between the integrated dendritic input and the axonic output of action potentials may constitute the classical clock for phase-locking of about  $q=12-14$  qubits in each of  $n=10^4$  neurons.

The enormous discrepancy between the number of quantum registers and that of neurons results from a decompression by one-to-one correlation of quantum states to classical information channels with  $f = 1/\text{neuron}$ . Remember, that this correlation warrants the freedom of the target neuron to choose among all quantum states by the "choice of channel" model (quantum output=channel input). The number of addressable quantum registers will increase by magnitudes if we consider multiple channel connections within this cluster of neurons. Accordingly, the brain is an ideal organ for decompression of quantum information. Stuart Hameroff and Roger Penrose calculated that a number of  $q=10^9-10^{10}$  entangled qubits may be sufficient for a moment of conscious experience [22]. If the "choice of channel is choice of neuron" selection implies that all combinations of neurons are allowed ( $f=2^n$ ), the number of entangled qubits may be equivalent to the number of neurons, e.g.,  $q=10^4$ . The advantage of this model is at hand: Consider a

string of three input channels, then the selection of active channels can be described by binary coding, e.g., 001, 011, 101 etc. corresponding to orthogonal ensemble states of a quantum system (e.g.,  $A$ ,  $A'$ ,  $A''$  etc.). This allows a very convenient one-to-one mapping of classical channels to quantum registers with each channel combination selective for a distinct oscillation phase. An interesting side-aspect of this approach is given by the possibility of digital programming of the classical channels and interfacing with conventional microcomputers which may be significant for the processing of input/output information in realistic quantum computers.

## 2.2. Compression of cbits for feeding of qubits

Most recently, efficient methods have been suggested for classical, e.g., fractal re-compression of communicated cbits onto the quantum registers of single neurons with logarithmic compression factor, e.g.,  $q=(n/2)^2$  [23]. In our example, two sets of  $10^4$  neurons may compress the information of  $q=10^8$  qubits which is very close to the number of entangled qubits suggested for conscious experience [22]. With an average time delay between dendritic input and axonic output reported to be 20 msec, the quantum clock of the neuronal cluster will switch with approximately 50 Hz. This is close to an overall frequency of 40 Hz found in the electrical brain activity. It should be mentioned, that a binding frequency of 40 Hz does not necessarily imply the participation of quantum processes revealing an average time of 20 msec for singlet evolution. Much smaller time scales for the input/output delay are possible if we include computation within the dendritic tree by coupled dendritic spines into the present model [24-26]. This will result in binding of network oscillations in the kHz-range which has also been reported for the coordination of action potentials in neuronal clusters [27,28]. It should be kept in mind, however, that there are several completely classical models for the generation of phase-locked oscillations in the human brain described in literature [29]. Quantum computation of the neuronal substrate could be revealed by oscillations in the MHz- or GHz-range which would be measured as random noise of sub-threshold potentials (see Part 3). Usually, the electrical noise of the neuronal membrane is referred to as thermal noise, however, with unclear significance for the generation of action potentials. It may involve neuronal quantum computation of a synergistic ordering parameter in a time frame escaping that for decoherence (for discussion see Part 3).

## 2.3. Superselection of classical orbits for non-local quantum evolution of a memory space

Imagine, that any target neuron may constitute an intersection point for column and row feeder neurons in a Hopfield network matrix [18-21]. In this case, the entire row could harmonically oscillate in resonance with columns differing by integer numbers of frequencies. The most important property of a self-organized neuronal network of quantum oscillations, however, is that the dynamic geometry of multiple connections

organizes and re-organizes the frequency response. This results from the tuning of the quantum oscillation by the simultaneous signal addition in one neuron. The signal propagation depends on the transmission time and hence the number of neurons and length of transmission cables. Suppose you have a ladder of  $n$  neurons with each neuron  $N$  as rung connected to two neurons on each side by a dendritic input and an axonic output stile. Now, there is a connection of  $N$  to  $N + 3$  by a second ladder acting on the first, and a connection of  $N$  to  $N + 2$  by a third ladder acting on the second one. The periodicity of the first ladder must be 6 by factorization with 2 times 3. This periodicity is the same in a three ladder construct with connection  $N$  to  $N + 2$  in the second and  $N + 3$  in the third ladder. This construct may store a oscillation periodicity independent of the construction history which is consistent with the memory space in our brain. Furthermore, activation of any ladder recalls the periodicity of the entire construct equivalent to associative memory. This simple model is even applicable to molecular automaton theory of interconnected microtubuli or ion channels [4,5,30,31]. A bundle of parallel microtubuli maybe interconnected by microtubuli associated proteins (MAPs) as input/output channels [31]. In the following section we will see that coupling of quantum-oscillations in a network of parallel ladders is not only useful for description of neuronal but also of sub-neuronal function.

The rows and column ladders in a Hopfield network may form closed loops (the first and the last neuron in a row or column are connected). This model constitutes a one-dimensional memory attractor given by the oscillation periodicity of a whole column or row. A column neuron may act as a feeder for tuning or pitching of the row resonance. Moreover, rows and columns can now form knotted and concatemerized loops. A string-like concatemer of loops can have an  $m$ -adic periodicity, e.g., a triadic string will show a periodicity of  $2 + 2N$ . A triadic periodicity will be selected from all oscillations in a network by synchronous input from two channels at neuron  $N = 1, 3, 7$  etc., or in general for all  $m$ -adic periodicity at connection neuron  $N^{(m-1)} + N^{(m-2)} + N^{(m-3)} + \dots + N + 1$ . Most recently, the model of  $m$ -adic periodicity in neural networks has been utilized to define the phase space of a memorized idea as just a string of  $m$ -adic neuronal coupling [32]. This string could be a row in a Hopfield network and would be activated by resonance with intersecting column strings. This puts the resonance model for loop knots with one common knotting neuron for each single loop into the frame of a neural network. The knotting neuron will have  $f$  communication channels with an equal number of feeder loops. Resonance will occur by synchronous input of all loops, that is to say at harmonic frequency relations. At this point it should be kept in mind that any selection of frequency relations in a neural network is intimately correlated with a superselection of harmonics for a particular quantum-oscillation in a single neuron.  $M$ -adic periodicity (in row strings) and harmonic frequency relations (from several knotted columns) will form a stabilized backup for associative memory with a potential of creativity: one idea in a row interconnected with other ideas by several columns will self-organize an orbit space for creativity.

Even higher convolution is possible. If we define a row in a planar Hopfield network layer as reference row then this row can be knotted with two other rows, one of a top and one of a bottom layer. This would introduce a third dimension of connection by columns orthogonal to stacked Hopfield layers. The frequency of the reference row is now in phase relation to other rows which introduces the possibility of a holographic memory in a linear series of (quantum)-resonators consistent with a holographic paradigm of brain function



[33-36]. If the phase differences are periodic the superposition generates an activation pattern in knotting neurons with three-dimensional fractal distribution [1,37,38]. From these models can already be seen that a particular quantum oscillation in a neuron can be non-locally embedded in a meme-like context dependence, as memory or as idea. The study presented in this Letter would most efficiently combine non-local quantum evolution of a Hilbert space with recent models for Darwinian selection of an idea space in the human mind by synchronization of action potentials in a neural network [39,40].

### Part 3: Localization of consciousness in a neuronal substrate

#### 3.1. Clonal selection of a configuration space for consciousness

The greatest miracle of human consciousness is its ability to experience a variety of different information (reducible) in a unified space-time (irreducible) [1]. Up to now, no conventional classical (by reduction) or quantum (by non-reduction) approach to describe a configuration space for consciousness has been convincingly successful. The most promising models based on quantum computation are likely to combine orthogonal (reducible) with non-orthogonal (irreducible) vector states for entanglement [2, 41]. However, the intricacies to correlate an imaginary orthogonal state with a reality experienced in the mind, and to technically realize non-orthogonal states in a neuronal substrate seriously impair any attempt to raise configuration spaces for consciousness above mere mathematical fiction. Nevertheless, the intimate experience of each of us to be conscious proves that this space exists. At first, a realistic attempt to define a configuration space for consciousness will most likely develop from a construction principle for non-orthogonal and orthogonal quantum states in a neuronal substrate which will be discussed in the next section. Irrespective of the nature of the biological substrate for quantum computation, the question to be approached is: How can an entangled state in one neuron communicate with those in other neurons in order to generate a common configuration space of consciousness? From this question it will be immediately clear that a solution is intimately linked to the problem of broadcasting of quantum states and purification of entanglement [42,43].

So far, the model presented here can be described as non-local Darwinian selection of local quantum states without true entanglement of different neurons. The effects observed remind of recent concepts in quantum computation referred to as quantum-nonlocality or EPR phenomenon without entanglement [44,45]. In the present study, each neuron re-generates an ensemble of orthogonal quantum states by its own output. Only the selection among given states is tuned by the classical input of other neurons. They function as "feeding environment" warranting the "survival of the fittest" couplings for selection of one clone of qubits on a single neuron. Non-resonant couplings will be extinguished. This would be sufficient if the entire configuration

space of consciousness can be found in the Hilbert space of entanglement in each of all neurons. There are results reported in literature that single neurons integrate a great deal of information perceived in consciousness [46]. However, if we claim that a configuration space is constituted by different neurons in a network we have to generate at least one non-orthogonal state by two neurons in order to fulfill the Peres-Horodecki inseparability criterion for entanglement [8-10, 42,43]. This is not trivial since an inseparable density operator generated by entanglement of one out of  $q$  qubits may result in as many  $2^{q-1}$  separable as inseparable states between two neurons. One communication channel for entanglement, even without knowing what its biological nature is, may thus serve as envelope function in order to shape the density matrix of  $2^{q-1}$  states in a single neuron. Most recently, it has been shown that a classical channel can be used in order to communicate one qubit for entanglement [47]. This may offer the intriguing possibility to combine different qualities of information, e.g., audiovisual, tactile, emotional etc. processed by different neurons in a common configuration space.

### 3.2. The quantum theater of consciousness in a single neuron

Even if the configuration space of consciousness can comprise system states on different neurons by communication of qubits via classical (e.g., axon/dendrite) channels, it is very likely that a great deal of the world consciously perceived by ourselves must be generated by the substrate of a single neuron. The uploading of information given by action potentials into an experience space of consciousness demands for a neuronal substrate which can dynamically interact with ion fluxes at the cell membrane. Taking the microtubule model as working hypothesis, it is hard to imagine which interface can be utilized in order to tune quantum effects in microtubules by the dynamical behavior of the neuronal membrane. Instead, it is suggested that the neuronal substrate of consciousness is constituted by the cell membrane itself. This assumption has often been rejected due to the vulnerability of the membrane to thermal noise which naturally induces rapid decoherence [6]. However, former considerations were focused on noisy opening/closing characteristics of ion channel proteins and neglected the elasticity characteristics of the lipid bilayer. At temperatures around 310 K (body temperature) lipid bilayers undergo sharp phase transitions into a liquid-crystalline phase [48,49]. They form a hexagonally packed lattice which is shielded from the aqueous environment by van der Waals forces between stacked hydrocarbon moieties of fatty acids or cholesterol. This bilayer is under a pressure of 400 Atm and undergoes dense packaging of the hydrophobic moieties with coherence lengths of more than 6 nm for model lattices of lipids surrounding ion channel proteins [49,50]. Dotting of the liquid-crystal membrane with ion channels will alter the surface pressure by opening (increase) or closing (decrease) of the protein complex [50]. Accordingly, action potentials and ion flux entail a lateral compression of the lipid bilayer and its increased compression may enhance the activation energy for channel opening (feedback-loop).

If we neglect for a moment the temperature of the system we will easily picture a striking homology of the neuronal cell membrane to quantum dot lattices with piezoelectric characteristics [51,52]. Opening of ion

channels by input of action potentials may entail a rotation of the hydrophobic lipid moieties (e.g., hydrocarbon chains of fatty acids) to a denser stacking with the result of altered dipole characteristics. In a condensed matter system this would lead to scattering effects by electron-phonon interactions [51,52]. If we rule out quantum effects in the membrane this behavior could be sufficiently described by a classical spin-glass system. However, if we consider a quantum coherent system as a priori assumption and suppose a stabilization by NQCC and multiple channel superselection as described before (Part 2.3.), the classical spin-glass will naturally turn into a quantum spin system. The rotational spin-spin interactions of the hydrocarbon chains could be directly translated into qubit rotations for entanglement. Accordingly, the molecular dynamics of the lipid bilayer corresponds to the translator concept in NQCC with reset of the lipid chain rotation being consistent with a Hadamard operation. The transformation of rectangular to triangular or hexagonal lattice structures by shearing forces may provide the necessary "square root swap" rotation by  $\pi/4$  for entanglement. Readout and conversion to classical information channels could be achieved by the lipid lattice-induced alteration of the ion channel conformation changing the ion flux across the membrane (back-action). The energy potential stored by the structural deformation of these proteins can then be used in order to restore entanglement of the lipids by acquirement of the original protein conformation (reset by elastic rebound effect). It is obvious, that the intrinsic molecular dynamics resulting from the topology of lipids/ion channel proteins comprises a set of pointer states which determine the interaction Hamiltonian with the system state of entangled hydrocarbon dipoles and induces the electron-phonon scattering effect of the lipid lattice. The critical question is: Can this dynamics or interaction be shielded from thermal decoherence?

If we only consider single molecule collisions as in Max Tegmark's paper on the quantum brain the answer is definitely "no" [6]. However, if we assume that NQCC is stabilized by synchronous input of multiple channels we may generate oscillations with a frequency for the interaction Hamiltonian which is higher than that for thermal decoherence. This model is consistent with the Frohlich concept for coherent vibrations in biological materials [53,54]. A simple calculation may give an idea about the time frame of these oscillations. A dendrite shaft is a rod-shaped branch of the dendritic tree which is densely studded with dendritic spines, the synaptic contact points for input potentials communicated from axons of other neurons. Suppose, that the synchronous frequency for a spike train in a neuronal cluster equals the the time difference between dendritic input and axonic output of a single neuron (approx. 20 msec or 50 Hz). If we assume that the regeneration time between two action potentials in a dendritic spine is about 1 msec and we calculate a synchronous input of  $10^4$  to  $10^5$  signals (= number of dendritic spines per tree)/msec we obtain an average time difference of  $10^{-8}$  sec between two signals arriving at the dendritic tree. On the basis of a neuronal signal velocity of about 100 m/sec we may easily see that this time is sufficient to couple the signals of two adjacent dendritic spines (100-500 nm apart) or ion channel proteins (10-100 nm apart) for coherent vibration. However, it is undoubtedly necessary to further improve this model in order to overcome thermal noise according to Tegmark's estimation of  $10^{-13}$  to  $10^{-20}$  sec decoherence time [6]. The following calculation will give a lower boundary for the frequency of oscillation modes. Provided that the movement of one charge by an ion channel corresponds to a changing potential of approximately  $U=5$  mV ( $= kT/ne$  with  $k$ =Boltzmann constant,  $T$ =temperature (310 K),  $e$ =unit charge,  $n$ =number of charges for induction of channel opening = 6)[55]. This corresponds to an energy  $E$  of  $8 \times 10^{-22}$  J ( $=Ue$ ) for acceleration of one charge. In this simplified case we would have to maintain an oscillation frequency of about  $10^{12}$  Hz ( $=E/h$ ) in order to escape thermal decoherence which is consistent with Frohlich's calculations for generation of coherent vibrations in biological substrates [53,54].

From these considerations it becomes clear that in addition to the synchronous input of  $10^4$ - $10^5$  synaptic signals/msec per neuron we have to claim a short range (about 1 nm) interaction in order to maintain a coherent coupling of vibrational nodes escaping thermal decoherence. This may be contributed by rotational interactions of the fatty acid chains or aromatic rings of phospho- or glycolipids or cholesterol arrays surrounding ion channel proteins. Assuming a hexagonal lattice with 6 nm coherence length as mentioned before the elementary module for entanglement would comprise a cell membrane area of about  $10^{-17}$  m<sup>2</sup> for a coherent cluster of one ion channel protein surrounded by approximately 100 lipid molecules. Due to the intrinsic rotational oscillation with a frequency of  $10^{12}$  Hz we will refer to this cluster as Frohlich module. Adjacent Frohlich modules interact with a frequency of  $10^8$  Hz resulting from synchronous multichannel input of  $10^4$ - $10^5$  dendritic action potentials/msec as discussed before (see also 1.3. and 2.1.). A coupling to the same number of channels in conjunction with an uploading time of 20 msec for each neuron gives a calculation of  $10^6$  input signals for one cycle. With appropriate compression as described before (Part 2.2.), a corresponding number of classical input channels can transmit the information of  $10^{11}$ - $10^{12}$  qubits into a single neuron. If we take a membrane area of  $10^{-17}$  m<sup>2</sup>/Frohlich module a complete neuronal cell surface would comprise about  $10^8$  ion channels surrounded by  $10^{10}$  lipid molecules in a coherent system. As can be seen by comparison of these numbers a "binding" frequency of approximately 50 Hz for a cluster of  $10^6$  connected neurons may feed a system of at least  $10^{10}$  entangled qubits for conscious experience. These may be generated in a single neuron, or  $10^4$  qubits in each of  $10^6$  neurons by shared entanglement in a common configuration space of the entire cluster. For all practical purpose, shared entanglement has advantage over compression of the experience space for consciousness into a single neuron. The reduction of the number of entangled qubits/neuron may be necessary due to i), limited surface array addressed by input channels (only dendritic tree or parts of it), ii) idleness of the ion channels leading to a considerable delay of observable oscillations below  $10^8$  Hz, iii) smaller number of input signals/msec compensated by intrinsic oscillation dynamics in order to restore entanglement of the Frohlich modules.

Recently, the significance of the dendritic tree for conscious experience has been described and ion channel characteristics in the kHz-range have been observed [24-28]. In particular, the electrical noise of the membrane has been correlated with quantum-fluctuations for conscious experience and discussed as way out of the time-scale for thermal decoherence [56] At this point the author wishes to emphasize that he is fully aware of the difficulties to maintain quantum coherence of even two particles at supercold temperatures. It is understood that the idea of  $10^{11}$  entangled qubits in the brain tissue will, with no doubt, provoke objection. Nevertheless, it is the principle of entanglement which accommodates with holistic phenomena unexplainable by classical physics. In future studies it will be shown how the intrinsic dynamics of the neuronal membrane may be correlated with "pixel" of conscious experience, the bits of consciousness ("conbits")..

**Acknowledgment:** The author wishes to thank Drs. Chris Nunn and Tal Mor for critical and helpful comments of the first version of this Letter which substantially improved the study.

## References

- [1] Bieberich, E. (1999) Structure in human consciousness: A fractal approach to the topology of the self perceiving an outer world in an inner space. Cogprints archive neuro/199902004.
- [2] Stapp, H. (1995) Why classical mechanics cannot naturally accommodate consciousness but quantum mechanics can. *Psyche* 2(5).
- [3] Beck, F. and Eccles, J.C. (1992) Quantum aspects of brain activity and the role of consciousness. *Proc. Natl. Acad. Sci. U.S.A.* 89, 11357-11361.
- [4] Jibu, M., Hagan, S., Hameroff, S.R., Pribram, K.H. and Yasue, K. (1994) Quantum optical coherence in cytoskeletal microtubules: implication for brain function. *Biosystems* 32, 195-209.
- [5] Mavromatos, N.E. and Nanopoulos, D.V. (1997) Microtubules: The neuron system of the neurons? LANL archive quant-ph/9702003.
- [6] Tegmark, M. (1999) The quantum brain. LANL archive quant-ph/9907009.
- [7] Cleve, R., Ekert, A., Henderson, L., Macchiavello, C. and Mosca, M. (1999) On quantum algorithms. LANL archive quant-ph/9903061.
- [8] Vedral, V. and Plenio, M.B. (1998) Basics of quantum computation. LANL archive quant-ph/9802065.
- [9] Hladky, b., Drobny, G. and Buzek, V. (1999) Quantum synthesis of arbitrary unitary operators. LANL archive quant-ph/9905049.
- [10] Barnes, J.P. and Warren, W.S.. (1999) Decoherence and programmable quantum computation. LANL archive quant-ph/9902084.
- [11] Benjamin, S.C. and Johnson, N.F. (1998) Cellular structures for computation in the quantum regime. LANL archive cond-mat/9808243.
- [12] DiVincenzo, D.P. (1998) Real and realistic quantum computers. *Nature* 393, 113-114.
- [13] Makhlin, Y., Schon, G., and Shnirman, A. (1999) Josephson-junction qubits with controlled couplings. *Nature* 398, 305-307.
- [14] Averin, D. V. (1999) Solid-state qubits under control. *Nature* 398, 748-749.

- [15] Amlani, I., Orlov, A.O., Toth, G., Bernstein, G.H., Lent, C.S., and Snider, G.L. (1999) Digital logic gate using quantum-dot cellular automata. *Science* 284, 289-291.
- [16] Burkard, G., Loss, D., and DiVincenzo, P. (1998) Coupled quantum dots as quantum gates. LANL archive quant-ph/9808026 v2.
- [17] Ashworth, M.C. (1997) Coherent state approach to quantum clocks. LANL archive quant-ph/9703029.
- [18] McKenna, T., Davis, J. and Zornetzer (1992) *Single neuron computation*. Academic Press, Inc., New York.
- [19] Haykin, S. (1994) *Neural networks. A comprehensive foundation*. Macmillan Coll. Publ. Comp., New York.
- [20] Muller, R., Reinhardt, J. and Strickland, M.T. (1995) *Neural networks*. Springer, New York.
- [21] Carpenter, G.A. and Grossberg, S. (1992) *Neural networks for vision and image processing*. MIT Press, Cambridge, MA.
- [22] Hameroff, S. and Penrose, R. "Conscious events in orchestrated space-time selections"
- <http://www.u.arizona.edu/~hameroff/hardfina.html>.
- [23] Bieberich, E. (1999) Generation of the self in a fractal quantum oscillator (FRAQTOR). Abstr. Flagstaff conference on "Quantum approaches to consciousness", Flagstaff, AZ..
- [24] Orpwood, R.D. (1994) A possible neural mechanism underlying consciousness based on the pattern processing capabilities of pyramidal neurons in the cerebral cortex. *J. Theor. Biol.* 169, 403-418.
- [25] Segev, I. (1998) Sound grounds for computing dendrites. *Nature* 393, 207-208.
- [26] Agmon-Snir, H. and Segev, I. (1993) Signal delay and input synchronization in passive dendrite structures. *J. Neurophys.* 70, 2066-2085.
- [27] deCharms, C.R. and Merzenich, M.M. (1996) Auditory cortical representation of sounds by the coordination of action-potential timing. *Nature* 381, 610-613.
- [28] Nakanishi, K. and Kukita, F. (1998) Functional synapses in synchronized bursting of neocortical neurons in culture. *Brain Res.* 795, 137-146.
- [29] Borisyuk, R.M. and Hoppensteadt, F.C. (1998) Memorizing and recalling spatial-temporal patterns in an oscillator model of the hippocampus. *BioSystems* 48, 3-10.
- [30] Akingbehin, K. (1995) A programmable interface to neuromolecular computing networks. *BioSystems* 35, 223-227.
- [31] Dayhoff, J., Hameroff, S., Lahoz-Beltra, R., and Swenberg, C.E. (1994) Cytoskeletal involvement in neuronal learning: a review. *Eur. Biophys. J.* 23, 79-93.

- [32] Dubischar, D., Gundlach, V.M., Steinkamp, O. and Khrennikov, A. (1999) A p-adic model for the process of thinking disturbed by physiological and information noise. *J. theor. Biol.* 197, 451-467.
- [33] Gabor, D. (1968) Improved holographic model of temporal recall. *Nature* 217, 1288-1289.
- [34] Longuet-Higgins, H.C. (1968) Holographic model of temporal recall. *Nature* 217, 104-105.
- [35] Pribram, K. (1982) What the fuss is all about, in "The holographic paradigm and other paradoxes" (Wiber, K, ed.) Shambala, London
- [36] Psaltis, D., Brady, D., Gu, X.-G., and Lin S. (1990) Holography in artificial neural networks. *Nature* 343, 325-330.
- [37] Bieberich, E. (1999) Neurochemical analysis of molecular interaction in neurochips may reveal internal mental states. Quantum-mind archive March 8<sup>th</sup>.
- [38] Bieberich, E. (1999) In search of a neuronal substrate of the human mind: new concepts from topological neurochemistry. *Cogprints archive neuro/199902006*.
- [39] Calvin, W. (1987) The brain as Darwin machine. *Nature* 330, 33-34.
- [40] Calvin, W. (1998) Competing for consciousness: A Darwinian mechanism at the appropriate level of explanation. *J. Consc. Stud.* 5, 389-404.
- [41] Fuchs, C.A. (1998) Just two nonorthogonal quantum states. LANL archive quant-ph/9810023.
- [42] Buzek, V., Vedral, V., Plenio, M.B., Knight, P.L. and Hillery, M. (1997) Broadcasting of entanglement via local copying. LANL archive qant-ph/9701028.
- [43] Horodecki, M., Horodecki, P. and Horodecki, R. (1998) Mixed-state entanglement and distillation: is there a "bound" entanglement in nature? LANL archive quant-ph/9801069
- [44] Bennett, C.H., DiVincenzo, D.P., Fuchs, C.A., Mor, T., Rains, E., Shor, P.W., Smolin, J.A. and Wootters, W.K. (1998) Quantum nonlocality without entanglement. LANL archive quant- ph/9804053 v4.
- [45] Horodecki, R., Horodecki, M. and Horodecki, P. (1998) Einstein-Podolsky-Rosen paradox without entanglement. LANL archive quant-ph/9811004
- [46] Miyashita, Y., Date, A., and Okuno, H. (1993) Configurational encoding of complex visual forms by single neurons of monkey temporal cortex. *Neuropsychologia* 31, 1119-1131.
- [47] Deutsch, D. and Hayden, P. (1999) Information flow in entangled quantum systems. LANL archive quant-ph/9906007.
- [48] Perillo, M.A., Scarsdale, N.J., Yu, R.K. and Maggio, B. (1994) Modulation by gangliosides of the lamellar-inverted micells (hexagonal II) phase transition in mixtures containing phosphatidylethanolamine and dioleoylglycerol. *Proc. Natl. Acad. Sci. U.S.A.* 91, 10019-10023.
- [49] Sperotto, M.M. and Mouritsen, O.G. (1991) Monte Carlo simulation studies of lipid order parameter

---

profiles near integral membrane proteins. Biophys. J. 59, 261-270.

[50] Knight, J. (1999) Ten, nine, eight...you're under. New Scientist 2178, 34-37.

[51] Khveshchenko, D.V. and Reizer, M. (1997) Piezoelectric electron-phonon interaction in impure semiconductors: 2D-electrons versus composite fermions. LANL archive cond-mat/9705254.

[52] Zanardi, P. and Rossi, F. (1998) Subdecoherent information encoding in a quantum-dot array. LANL archive quant-ph/9808036.

[53] Frohlich, H. (1970) Long range coherence and the action of enzymes. Nature 228, 1092-1093.

[54] Frohlich, H. (1975) The extraordinary dielectric properties of biological materials and the action of enzymes. Proc. Natl. Acad. Sci. U.S.A. 72, 4211-4215.

[55] Green, M. (1998) A resonance model gives the response to membrane potential for an ion channel. J. theor. Biol. 193, 475-483.

[56] Lisewski, A.M. (1999) Can all neurobiological processes be described by classical physics? LANL archive quant-ph/9907052.

.

